01 / ENGINEERING & MATHEMATICAL THEORY

2D rotations and matrices

EX29 · Algebraic foundations, geometry and physical applications

Reading order
Read this document before the Python Reference Guide and the Exercise Guide. Mathematical concepts come first, followed by geometric and engineering interpretations. No data files or automated tests are needed.

1. What does it mean to rotate in the plane?

A planar rotation is a **linear transformation** that changes a vector's orientation about a fixed origin without changing its length. Assume a right-handed Cartesian frame, column vectors, and counterclockwise positive angles.

A two-component vector is written as

$$\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix} = v_x \hat{\imath} + v_y \hat{\jmath}.$$

Here $\hat{\imath}$ and $\hat{\jmath}$ are the Cartesian unit vectors along x and y. They should not be confused with the imaginary unit from complex numbers.

2. Algebraic derivation of the rotation matrix

Start with the Cartesian basis vectors $\mathbf{e}_x = (1, 0)^T$ and $\mathbf{e}_y = (0, 1)^T$. Rotating these by θ counterclockwise gives

$$R(\theta)\mathbf{e}_x = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \qquad R(\theta)\mathbf{e}_y = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}.$$

Place **these images as columns** of the matrix, following the same principle used with actuator directions in EX28:

$$\boxed{R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}}.$$

Matrix-vector multiplication yields

$$\mathbf{v}_{\text{rot}} = R(\theta)\mathbf{v} = \begin{pmatrix} v_x\cos\theta - v_y\sin\theta \\ v_x\sin\theta + v_y\cos\theta \end{pmatrix}.$$

3. Worked mathematical example: 90-degree rotation

Let $\mathbf{v} = (2, 1)^T$ and $\theta = 90°$. Since $\cos 90° = 0$ and $\sin 90° = 1$:

$$R(90°) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Apply the matrix:

$$\mathbf{v}_{\text{rot}} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

Check the vector norms:

$$\|\mathbf{v}\|_2 = \sqrt{2^2 + 1^2} = \sqrt{5}, \qquad \|\mathbf{v}_{\text{rot}}\|_2 = \sqrt{(-1)^2 + 2^2} = \sqrt{5}.$$

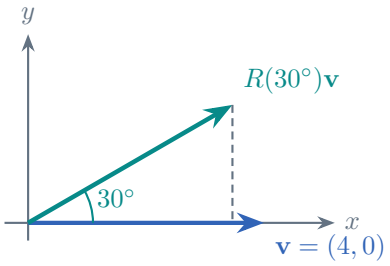
Orientation changes, but length does not.

4. Worked mathematical example: 30-degree rotation

Let $\mathbf{q} = (4, 0)^T$. Since $\cos 30^\circ = \sqrt{3}/2$ and $\sin 30^\circ = 1/2$:

$$R(30^\circ)\mathbf{q} = \begin{pmatrix} 4\cos 30^\circ \\ 4\sin 30^\circ \end{pmatrix} = \begin{pmatrix} 2\sqrt{3} \\ 2 \end{pmatrix}.$$

The first component is approximately 3.464 and the second is 2.



5. Important rotation properties

The columns of $R(\theta)$ have unit norm and are mutually perpendicular; thus R is an **orthogonal matrix**:

$$R(\theta)^T R(\theta) = I, \qquad R(\theta)^{-1} = R(\theta)^T = R(-\theta).$$

Moreover, $\det R(\theta) = 1$. Rotations preserve the norm and dot product:

$$\|R\mathbf{v}\|_2 = \|\mathbf{v}\|_2, \qquad (R\mathbf{u}) \cdot (R\mathbf{v}) = \mathbf{u} \cdot \mathbf{v}.$$

In 2D, with rotations about the same fixed axes, compositions add angles:

$$R(\alpha)R(\beta) = R(\alpha + \beta).$$

Do not assume this planar result applies to rotations about distinct 3D axes, where operation order generally matters.

6. Rotating a vector versus rotating the axes

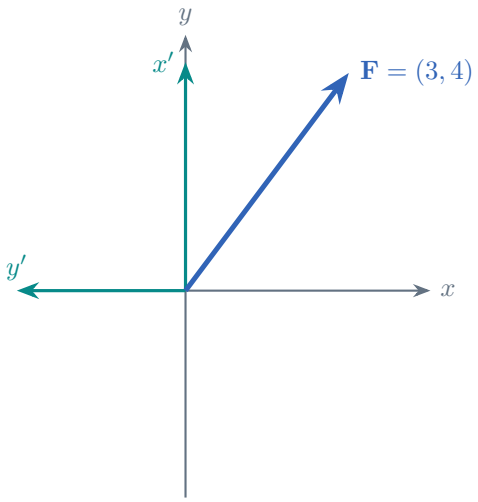
Active rotation: the coordinate system remains fixed while the vector itself rotates. The operation is $\mathbf{v}_{\text{new}} = R(\theta)\mathbf{v}_{\text{old}}$.

Passive change of basis: the physical vector stays fixed. Its components are expressed in new axes rotated by $+\theta$ relative to the original axes:

$$\boxed{\mathbf{v}_{\text{rotated axes}} = R(\theta)^T \mathbf{v}_{\text{global}} = R(-\theta)\mathbf{v}_{\text{global}}}.$$

For example, let $\mathbf{F}_{\text{global}} = (3, 4)^T$ and rotate the *axes* by $+90^\circ$:

$$\mathbf{F}_{\text{new axes}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}.$$



The physical vector stays fixed; only the axes used to express its components change.

The transpose appears because the **unit vectors of the rotated axes** are the columns of R expressed in the original basis. The transpose computes projections onto the new axes, exactly as in the supplementary EX28 transformation note.

7. Physical application: rotating a force

An initially horizontal force of magnitude 10 N is $\mathbf{F} = (10, 0)^T$ N. If its direction rotates physically through 30°:

$$\mathbf{F}' = R(30^\circ)\mathbf{F} = \begin{pmatrix} 5\sqrt{3} \\ 5 \end{pmatrix} \text{N} \approx \begin{pmatrix} 8.660 \\ 5.000 \end{pmatrix} \text{N}.$$

The force magnitude remains 10 N. The rotation matrix is dimensionless; the force units are unchanged.

8. Python connection and checks

NumPy trigonometric functions expect **radians**. Convert the angle with `np.deg2rad()`, calculate R using `np.sin()` and `np.cos()`, and use `@` for matrix-vector multiplication.

Essential distinction
Active: rotate the vector $\Rightarrow R\mathbf{v}$. **Passive:** express a fixed vector in rotated axes $\Rightarrow R^T\mathbf{v}$. Always specify which basis the matrix columns represent and in which direction the transformation acts.